

ISM Note 9

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1. Star formation (gravitational collapse)

We consider a gas cloud (protostar) with mass M and radius R . When gravity dominates over thermal pressure, the cloud will collapse.

$$\frac{GMm}{R^2} \gtrsim k_B T$$

where m is the mean mass per particle. This gives us the Jeans mass:

$$M_J \sim \left(\frac{k_B T}{Gm} \right)^{3/2} \rho^{-1/2}$$

where ρ is the density of the cloud. **This is the minimum mass required for a cloud to collapse under its own gravity.**

So the question is: For ISM, the temperature and density are commonly constants or around some typical values, so the Jeans mass is also a constant. Then why do we have stars with different masses? The answer is given by the so-called **fragmentation**. When a cloud collapses, its density will increase, which will decrease the Jeans mass. This allows smaller fragments of the cloud to collapse and form stars with different masses. **This fragmentation gives rise to the initial mass function (IMF) of stars.**

Another question is: At which point will the collapse stop? One important factor is the **radiation pressure** brought by the nuclear reactions. When collapse happens, the increasing density means the cloud will become **more optically thick** and it will be harder for it to lose energy through radiation. This will lead to a higher temperature and thus increase the Jeans mass. Finally, the collapse will stop when the thermal pressure is comparable to the gravitational pressure.

We have several timescales of the collapse process. The **free-fall timescale** is the time it takes for a cloud to collapse under its own gravity, which is given by:

$$t_{\text{ff}} \sim \sqrt{\frac{1}{G\rho}}$$

The free-fall timescale can be a good measure of the strength of gravity.

Cloud Collapse (Angular Momentum)

Consider a galaxy with Keplerian rotation.

$$\Omega \sim \frac{1}{r^2}$$

This means the inner part will rotate faster than the outer part, which will transfer angular momentum from the inner part to the outer part.

$$\delta\Omega_{\text{cloud}} \sim \frac{d\Omega}{dR} \delta R \sim \frac{\Omega(R)}{R} r_{\text{cloud}}$$

If angular momentum is conserved, the centrifugal force will balance gravity, and the cloud will form a disk not a star, **it will never collapse**. So we need to transfer the angular momentum out of the cloud. This can be done through magnetic fields.

Consider the flux freezing condition, the magnetic field lines will be dragged by the cloud, and we have

$$B \propto r^{-2}$$

For the sun, however by observations, we have

$$B_{\odot} \ll B_{\text{ISM}} \left(\frac{R_{\text{ISM}}}{R_{\odot}} \right)^2$$

The magnetic field of the sun is about 10^3 Gauss, which is much smaller than the ISM magnetic field. The possible reason might be the **ambipolar diffuse**. The B-field line will "slip" through the neutral component, so the magnetic field will decrease compared to the ideal case.

Vorticity Equation

The vorticity equation is another way to write down the momentum equation in terms of the vorticity, which describes the evolution of the vorticity. This is given by

$$\begin{aligned}\frac{\partial \vec{\omega}}{\partial t} = & -(\vec{v} \cdot \nabla) \vec{\omega} + (\vec{\omega} \cdot \nabla) \vec{v} - \vec{\omega}(\nabla \cdot \vec{v}) \\ & + \frac{1}{\rho^2} \nabla \rho \times \nabla(p + \frac{B^2}{8\pi}) + \frac{1}{4\pi\rho^2} \nabla \rho \times (\vec{B} \cdot \nabla) \vec{B}\end{aligned}$$

The first term is the **advection term**, which describes the transport of vorticity by the flow, if you do the volume integral of the whole space, this term will vanish.

The second term is the **stretching term**, which describes the stretching of vorticity by the flow.

The third term is the **compression term**, which has a negative sign before it. If we only have the first three terms, it is called **barotropic flow**. The fourth term is called **baroclinic flow**, which can generate vorticity, while the last term describes the magnetion tension to refrain the generation of vorticity.

Meaning of Sound Speed

We can define the so-called Mach number as

$$\mathcal{M} = \frac{v}{c_s}$$

We can describe several "information flow" cases with different Mach numbers with the following figure:

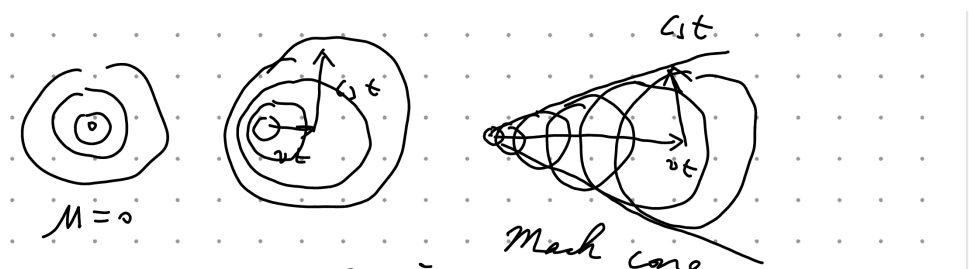


Fig 1. Flow with different Mach numbers.

The supersonic case will create a **shock**, and when the shock happens, we can see some **discontinuities**. To see it better, let's look at the steady state of the force equation, which is given by

$$\rho(\vec{v} \cdot \nabla)\vec{v} = -\nabla P + \rho\vec{f}$$

If we do some dimensional analysis and make the gradient P term which acts as the inertia to have dimension 1, we can get the dimension of the LHS term is $\frac{v^2}{c_s^2}$, and the dimension of the last term in RHS is $\frac{U_f}{c_s^2}$. So if $\mathcal{M} \gg 1$, the first term will dominate, you drive the flow so fast that there is no time for the flow itself to relax, and the flow will have jumps and discontinuities.

2. Stellar Winds (No ISM Interaction)

First of all, we have the mass conservation condition, which is given by

$$\rho v r^2 = \text{constant}$$

We can do the logarithmic derivative of the above equation, and we can get

$$\frac{1}{v} \frac{dv}{dr} + \frac{1}{\rho} \frac{d\rho}{dr} + \frac{2}{r} = 0$$

Then we can write down the momentum equation, which is given by

$$\rho v \frac{dv}{dr} + \frac{dP}{dr} = \rho \frac{GM}{r^2}$$

With $\frac{dP}{dr} = c_s^2 \frac{d\rho}{dr}$, we can plug in the result from the mass conservation condition, and then can get

$$\left(v - \frac{c_s^2}{v}\right) \frac{dv}{dr} = \frac{2c_s^2}{r} - \frac{GM}{r^2}$$

We don't have an analytic solution for the above equation, but we can do some analysis for some scenarios.

First, when $v = c_s$, since the derivative of the velocity will not blow up, we can get the **critical point**, which is given by

$$r_s = \frac{GM}{2c_s^2}$$

This is the so-called **sonic point**, which is the point where the flow speed equals the sound speed.

Second, when $r > r_s$, we assume the wind is **isothermal**, so we have $c_s^2 = \text{constant}$, and when $r \gg r_s$, we can get the asymptotic solution as

$$v^2(r) \sim 4c_s^2 \ln \frac{r}{r_s}$$

We can see from this that outside the sonic point, the flow becomes supersonic but the Mach number will only be a few and not by too much.

Third, when $r < r_s$, the gravity dominates, we can derive a solution with $\frac{dv}{dr} > 0$ and $v^2 < c_s^2$. And the asymptotic equation is given by

$$\frac{c_s^2}{v} \frac{dv}{dr} = -\frac{GM}{r^2}$$

And it looks like gravity drives wind out.

Now we consider a case where the stellar wind penetrates through the inhomogeneous ISM. When the stellar wind is blown out from the star, it will interact with the cloud and have something called "wind-cloud interaction". This will create turbulence and may affect the evolution of the cloud.

Inflow(Spherical Accretion, No Angular Momentum)

Basically the inflow is triggered by gravity. The energy changing rate is given by

$$\dot{E}_g \sim \frac{GM\dot{M}}{r}$$

where $\dot{M}$ is the mass accretion rate. And the luminosity of the accretion is given by

$$L_{\text{acc}} \sim \epsilon \dot{E}_g \sim \epsilon \frac{GM\dot{M}}{r}$$

where ϵ is the radiation efficiency. From this we can compute the radiation temperature as

$$\sigma T^4 \sim \frac{L_{\text{acc}}}{4\pi r^2}$$

For example, the stellar mass BH has $T \sim 10^7$ K, and the supermassive BH has $T \sim 10^{4-5}$ K.

Bondi Accretion (1D Spherical)

The Bondi accretion is a smooth and adiabatic inflow. The mass accretion rate is given by

$$\dot{M} = 4\pi r^2 \rho(r) v(r)$$

With $P \propto \rho^\gamma$ and $c_s^2 = \frac{\gamma P}{\rho}$, we can get the following equation:

$$\frac{1}{\rho} \frac{dP}{dr} = \frac{\gamma}{\gamma - 1} \frac{d}{dr} \left(\frac{P}{\rho} \right)$$

Then we can plug in the above equation into the momentum equation, and we can get

$$\frac{v^2}{2} + \frac{1}{\gamma - 1} c_s^2 - \frac{GM}{r} = \text{constant}$$

When $r \rightarrow \infty$, we have $v \rightarrow 0$ and $c_s \rightarrow c_{s,\infty}$, so we can get the constant in the RHS is given by

$$\frac{1}{\gamma - 1} c_{s,\infty}^2$$

And when $r \rightarrow r_s$, we have $v = c_s$, then we can get

$$\frac{1}{2} c_s^2(r_s) + \frac{1}{\gamma - 1} c_s^2(r_s) + 2c_s^2(r_s) = \text{constant}$$

Combining the above two equations, we can get the relation between the sound speed at the sonic point and the sound speed at infinity, which is given by

$$c_s^2(r_s) = \frac{2}{5 - 3\gamma} c_{s,\infty}^2$$

And plug this into $P \propto \rho^\gamma$ and $c_s^2 = \frac{\gamma P}{\rho}$, we can get

$$\rho(r_s) \sim \rho_\infty \left(\frac{c_s(r_s)}{c_{s,\infty}} \right)^{\frac{2}{\gamma-1}}$$

Finally we can get the **Bondi accretion rate**, which is given by

$$\dot{M} = 4\pi r_s^2 \rho(\infty) c_{s,\infty} \left(\frac{c_s(r_s)}{c_{s,\infty}} \right)^{\frac{\gamma+1}{\gamma-1}}$$

From this we can get

$$\dot{M} \propto M^2 \rho_\infty c_{s,\infty}^{-3}$$

And the **Bondi radius** is given by

$$r_B = \frac{GM}{c_{s,\infty}^2}$$

It shows the region where gas can be captured by gravity. For the supermassive BH, the Bondi radius is about 10 – 100 pc. **When the gas is within the Bondi radius, it will be captured by the gravity of the BH and eventually accrete onto the BH forming an accretion disk.**

Eddington Luminosity

We have a definite gravitational radius (Bondi radius) $r_B = \frac{GM}{c_{s,\infty}^2}$. The luminosity is thus given by

$$L = \epsilon \frac{GM\dot{M}}{r_B} \frac{r_B}{r} = \epsilon \frac{r_B}{r} \dot{M} c_{s,\infty}^2$$

For the bright quasars, the luminosity can be as high as 10^{46} erg/s, and the mass accretion rate can be as high as $0.35 M_{\odot}/\text{yr}$.

So the question is: what's the luminosity at which accretion flow stops?

The answer of this question is left to the next lecture.